KaTeX Cheat Sheet

A quick reference for writing mathematical expressions using **KaTeX** in Markdown.

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1. Basic Math Modes

Inline Math

Wrap expressions with single dollar signs.

Example: To write P(X=1), use: $P(X = 1)$

Display Math (Centered)

Use double `$$` on separate lines for centered equations.

```
 $$
\pi_{t+1} = \pi_t P
$$
```

Result:

\[\pi_{t+1} = \pi_t P\]

---

2. Greek Letters

Common letters: `\alpha`, `\beta`, `\gamma`, `\delta`, `\lambda`, `\pi`, `\theta`, `\sigma`, `\omega`

Examples:

- Type `\alpha` renders as α
- Type `\beta` renders as β
- Type `\gamma` renders as γ
- Type `\pi_t` renders as π_t
- Type `\theta` renders as θ

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3. Superscripts and Subscripts

What To Type	Result
x^2	x^2
x_t	x_t
x_{t+1}	x_{t+1}
P^t	P^t
x_i^2	x_i^2

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4. Fractions

Type: `\frac{numerator}{denominator}`

Examples:

- Type `\frac{q}{p}` renders as $\frac{q}{p}$
- Type `\frac{x+1}{y-2}` renders as $\frac{x+1}{y-2}$

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5. Square Roots

Type: `\sqrt{x}` or `\sqrt[n]{x}` for nth root

Examples:

- Type `\sqrt{x}` renders as $\sqrt{x}$
- Type `\sqrt{x^2 + y^2}` renders as $\sqrt{x^2 + y^2}$
- Type `\sqrt[3]{27}` renders as $\sqrt[3]{27}$

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6. Matrices (Very Useful for Markov Chains)

Type:

```
 \begin{bmatrix}
a & b \\
c & d
\end{bmatrix}
```

Result:

\[P = \begin{bmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{bmatrix}\]

Other styles:

- `pmatrix` (parentheses) renders as $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$
- `bmatrix` (brackets) renders as $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$
- `vmatrix` (vertical bars) renders as $\begin{vmatrix} a & b \\ c & d \end{vmatrix}$

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7. Probability Notation

Code	Result
P(A \mid B)	$P(A \mid B)$
P(X_{t+1}=j \mid X_t=i)	$P(X_{t+1} = j \mid X_t = i)$

8. Expectation

Code	Result
E[X]	$E[X]$
\mathbb{E}[X]	$\mathbb{E}[X]$
\mathbb{E}[X \mid Y]	$\mathbb{E}[X \mid Y]$

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9. Summation & Products

Code	Result
\sum_{i=1}^{n} x_i	$\sum_{i=1}^{n} x_i$
\prod_{i=1}^{n} x_i	$\prod_{i=1}^{n} x_i$

Display mode (centered):

\[\sum_{i=1}^{n} x_i\]

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10. Integrals

Code	Result
\int_a^b f(x)\,dx	$\int_a^b f(x)\,dx$
\iint_D f(x,y)\,dA	$\iint_D f(x,y)\,dA$

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11. Aligned Equations

For step-by-step derivations:

```
 \begin{aligned}
x &= y + 2 \\
  &= 3 + 2 \\
  &= 5
\end{aligned}
```

Result:

\[\begin{aligned} x &= y + 2 \\ &= 3 + 2 \\ &= 5 \end{aligned}\]

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12. Vectors

Code	Result
\vec{x}	$\vec{x}$
\mathbf{x}	$\mathbf{x}$
\mathbf{v} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}	$\mathbf{v} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$

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13. Sets

Code	Result
\{1,2,3\}	$\{1, 2, 3\}$
S = \{0,1,2,3\}	$S = \{0, 1, 2, 3\}$

Code	Result
A \cup B	$A \cup B$
A \cap B	$A \cap B$
A \subseteq B	$A \subseteq B$
x \in S	$x \in S$
x \notin S	$x \notin S$

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14. Markov Chain Notation

Transition matrix:

\[P = \begin{bmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{bmatrix}\]

State distribution:
Type `\pi_t = [0.2, 0.8]` renders as $\pi_t = [0.2, 0.8]$

Evolution:
Type `\pi_{t+1} = \pi_t P` renders as $\pi_{t+1} = \pi_t P$

Steady state:
Type `\pi = \pi P` renders as $\pi = \pi P$

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15. Logical Symbols

Code	Result
A \Rightarrow B	$A \Rightarrow B$
A \Leftarrow B	$A \Leftarrow B$

Code	Result
A \iff B	$A \iff B$
\forall x	$\forall x$
\exists x	$\exists x$
x \neq y	$x \neq y$
x \leq y	$x \leq y$
x \geq y	$x \geq y$
x \approx y	$x \approx y$

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16. Limits & Infinity

Code	Result
\lim_{x \to \infty} f(x)	$\lim_{x \to \infty} f(x)$
\lim_{n \to 0} \frac{1}{n}	$\lim_{n \to 0} \frac{1}{n}$
\infty	∞

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17. Norms & Absolute Values

Code	Result
\\|x\\|	$\|x\|$

Code	Result
\\|\mathbf{v}\\|_2	$\|\mathbf{v}\|_2$
\\|x - y\\|	$\|x - y\|$

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18. Spacing

Use for better readability:

- Type `\,` for small space: `f(x)\,dx` renders as $f(x)\,dx$
- Type `\quad` for medium space: renders as $a \quad b$
- Type `\qquad` for large space: renders as $a \qquad b$

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19. Quick Reference

Operation	Code
Fraction	\frac{a}{b}
Subscript	x_i
Superscript	x^2
Square root	\sqrt{x}
Sum	\sum_{i=1}^{n}
Product	\prod_{i=1}^{n}
Integral	\int_a^b
Infinity	\infty

Operation	Code
Pi	\pi
Alpha	\alpha

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20. Best Practices

1. **Inline Math:** Use single `$...$` for math within text
2. **Display Math:** Use `$$...$$` on separate lines for standalone equations
3. **Grouping:** Always use braces for multi-char subscripts: `x_{t+1}` not `x_t+1`
4. **Integrals:** Add spacing before dx: `\int f(x)\,dx`
5. **Conditionals:** Use `\mid` for probability, not `|`
6. **Common Mistake:**
   - Wrong: `x_t+1` renders as $x_t + 1$
   - Right: `x_{t+1}` renders as x_{t+1}

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21. Official Documentation

Full reference: https://katex.org/docs/supported.html

End of cheat sheet.